

Debug Lab Frontend Challenge

Two responsive Debug Lab pages built for the front-end developer technical assessment:

- [software-development.html](#)
- [stem-workshop.html](#)

The project also includes a lightweight local [index.html](#) so the shared navigation can be tested end to end.

How to Run

Open [index.html](#) directly in a browser, then use the navigation to visit the Software Development and STEM Workshop pages.

No build step is required.

Tech Stack

- Semantic HTML5
- Modern CSS with custom properties, Flexbox, Grid, [gap](#), and [clamp\(\)](#)
- Vanilla JavaScript for mobile navigation and active link state
- Google Fonts: Outfit
- Material Symbols for icon consistency with the Debug Lab landing page

I chose plain HTML, CSS, and JavaScript because the assessment pages are static, and the brief asked for clean, maintainable frontend code without unnecessary framework overhead.

Key Decisions

- The official PDF was treated as the source of truth for scope, content requirements, accessibility, responsiveness, and scoring criteria.
- The navigation follows the actual Debug Lab landing page structure: logo lockup, Programs dropdown, Events dropdown, Marketplace, Wall of Fame, Leaderboard, Coaches link, Student button, and mobile drawer.
- The implementation recreates the design in original code. It does not copy the Debug Lab source code or use Tailwind/Bootstrap.
- The color system is constrained to the requested palette: [#000000](#), [#F24405](#), [#F25C05](#), [#F28705](#), and [#F2F0E4](#).
- Reusable page patterns were centralized in [assets/css/style.css](#) and [assets/css/navbar.css](#) so both pages feel like one product.

Accessibility Notes

- Pages use semantic landmarks: [header](#), [nav](#), [main](#), [section](#), [article](#), and [footer](#).
- The mobile menu exposes [aria-expanded](#), supports Escape to close, and closes when a link or overlay is selected.

- Current page links are marked with `aria-current="page"` by JavaScript.
- Focus states are visible and keyboard-accessible.

What I Would Improve With More Time

- Replace placeholder links for Marketplace, Wall of Fame, Leaderboard, Coaches, and Student with confirmed production URLs.
- Add locally hosted optimized image assets if provided by Debug Lab.
- Add a small shared HTML include/build step to remove repeated header and footer markup while keeping the deployed output static.
- Run a full browser/device QA matrix and Lighthouse accessibility/performance checks before final submission.